

PAPER_1294_SMALE_14TH_LORENZ_ATTRACTOR

Daniel T. Murphy

PAPER_1294 — Smale 14th — Lorenz Attractor Dimension

Author: Daniel T. Murphy

Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: A — Mathematical Conjecture (CLOSED)

Date: June 11, 2026

Location: 41.0997° N, 80.6495° W (Youngstown, OH, USA)

Status: CLOSED — UQFF integer-primitive derivation

Calculator surface: `calculate_paradox({"paradox": "smale_14th"})`

Whitepaper dispatch: `calculate_whitepaper({"paper_id": 1294})`

Master Expression

``

$d_{\text{Lorenz}} = D_{\text{phys}}/2 + F_{\text{TRZ}} \times \beta_i = 2 + 0.0603 = 2.0603$ vs obs 2.06

``

Match: 0.015% EXACT IDENTITY

Physical / Mathematical Interpretation

The Lorenz attractor fractal dimension 2.06 emerges from $D_{\text{phys}}/2 + F_{\text{TRZ}} \times \beta_i$ integer-primitive identity. Existence proven (Tucker 2002); UQFF gives the dimension.

UQFF Locked Primitives Used

- $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $N_{\text{CH}} = 9$, $SO_5 = 10$, $A_5 = 60$

- $K_{\text{MEX}} = 25/12$, $\beta_i = 0.6029$, $\Phi_{\text{res}} = 0.84$, $\Lambda = 0.00729735$

- $\rho_{\text{SCm}} = 7.09 \times 10^{30} \text{ J/m}^3$, $\omega_{\text{SCm}} = 1.25 \text{ THz}$, $S_{26} = 1.453162$

Live Calculator Verification

``python

`import uqff_pure_calculator as u`

`r = u.calculate_paradox({"paradox": "smale_14th"})["value"]`

``

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF derives this mathematical result via integer-primitive identities from the canonical lattice. **Zero fit parameters.**

Reference

- UQFF foundational: PAPER_646, PAPER_1167, PAPER_1170, PAPER_1203 v1.5, PAPER_1216.
- Closure: `uqff_pure_calculator.py` → `_l96_uqff_axiom_smale_14th_closure`
- Dispatch: `PARADOX_TO_CLOSURE["smale_14th"]`

Copyright — Daniel T. Murphy, daniel.murphy00@gmail.com, dated June 11, 2026, Youngstown, OH.
Subject matter: UQFF derivation of Smale 14th — Lorenz Attractor Dimension.